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Impact of terminal heat stress on wheat yield in India and options for adaptation

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ABSTRACT

The effect of climate change is being observed in India in the form of shorter winter coupled with commencement of summer quite earlier than normal. The proximity of equator and late sowing of wheat (due to late harvesting of rice) exposes the wheat crop (*Triticum aestivum* L.) to high temperature stress during grain filling stage leading to terminal heat stress in the crop and reduced yield. There are limited studies done till date in India to assess the long term impact of climate variability on wheat yield and to develop adaptation strategies to reduce its negative impact. Therefore, in this study, simulation model (InfoCrop) and field experiment was used to assess the impacts of terminal heat stress on growth and yield of wheat as well as to identify adaptation strategies. Simulated results showed that terminal heat stress will reduce wheat yield by 18.1%, 16.1% and 11.1%, respectively in present, 2020 and 2050 scenarios. Advancement in sowing date, application of additional dose of nitrogen and irrigation at grain filling stage were found suitable options for preventing yield loss. Among various combinations of adaptation options, early sowing by 10 days from recommended sowing date with 30 kg ha^{-1} additional nitrogen fertilizer and one additional irrigation at grain filling stage was found most suitable. Adaptation of these strategies will help in reducing impact of heat stress by 7.5, 6.4 and 9% respectively in present, 2020 and 2050 heat stressed scenarios.

1. Introduction

Agriculture is the key economic sector in India for ensuring food, employment, and income security. It provides employment to around 66% of the workforce in India and represents 14% of India's gross domestic product. Rice-wheat cropping is the major cropping system grown in rotation on 13.5 Mha in the Indo-Gangetic plains (IGP) (Pathak et al., 2003; Panigrahy et al., 2011) producing about 50% of the total food grain and feeding 40% of India population (Gupta et al., 2016). India shares approximately 12% of world wheat production. In India, its production has risen from 6.5 Mt. in 1960 to 97.1 Mt. in 2017–18 (Directorate of Economics and Statistics, 2018). In the last few decades climate change is changing our abode in which our agricultural crops and different agronomic practices are changing rapidly (Rosenzweig et al., 2014). As per the projection South Asia will be deficient in 22 Mt. of food-grain production by 2030. Now this rice-wheat cropping system has started showing declining marginal yields, groundwater depletion, soil deterioration and heat stress problem in last decade. Rise in temperature is one of the most important concerns as metabolic processes that eventually impact the agricultural production is governed by temperature (Teixeira et al., 2013). In future increase in temperature will negatively affect agricultural production and more serious effect will be from short episodes of extremely high temperatures, or occurrence of 'heat stress. This effect is being observed in India especially in wheat crop since last decade. India is witnessing decrease in length of winter period and early commencement of significantly higher temperature than normal trend. Early commencements of significantly higher temperature coincide with wheat grain filling stage especially in Indo-Gangetic plains leading to terminal heat stress and reduction in yield. Indian proximity to the equator and late sowing of wheat exposes the crop to high temperature during grain filling stage limiting its production and productivity especially in Indo-Gangetic plains. Reason being the, proximity to the equator and late sowing of wheat in India which exposes the crop to high temperature

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Table 1
Model input requirement and output.

Model Input data requirement	Model output
Daily weather data (maximum temperature, minimum temperature, sunshine hours, daily mean rainfall), soil data (soil depth, slope, soil texture, pH, organic carbon, bulk density, soil moisture, water holding capacity of soil, soil ammoniacal and nitrate nitrogen content.), varietal information (growing degree days, temperature requirement for different phenological stages, specific leaf area, maximum relative growth rate, maximum potential test weight and radiation-use efficiency), crop management data (seed rate, sowing dates, crop duration, irrigation amount and dates, fertilizer requirement).	Total dry matter, crop yield, leaf area index, weight of plant parts (root, stem and leaf), grain number, crop N uptake, soil organic carbon, soil N status, soil water content, evapotranspiration, soil N loss, methane emission from soil etc.

during grain filling stage (Joshi et al., 2007). Terminal heat stress in wheat occurs when mean temperature during grain filling stage goes above 31 °C. The IPCC (2014) projected that for Indian region, temperature will increase by 0.7–2.0 °C by 2030s and 3.3–4.8 °C by 2080s. The increase may be more in north India and during the *rabi* season (November to March). With a rise of temperature by 0.5–1.56 °C by 2080–2100, adverse impact on food production will occur which will cause a loss of 10–40% in food grain production in India (Parry et al., 2004; IPCC, 2007). High temperature causes adverse impact on wheat physiology restricting growth and yield. At flowering stage, heat stress causes sterility of pollen and anther leading to underdeveloped embryo which in turn reduces grain number, while heat stress during grain filling stage leads to reduced grain filling rate and in turn reduces grain weight and overall yield (Mondal et al., 2013). In a study it was found that even a rise of 1 °C in the mean temperature in month of March–-April leads to reduction in the duration of wheat crop by seven days and yield by about 400 kg per hectare (Singh et al., 2011). Similarly, another analysis carried out for IGP indicates that with each 1 °C temperature rise there is risk that we may lose around 4–5 Mt. of wheat production (Aggarwal, 2008). An increase of 3 °C in temperature, the loss is projected to be 19 Mt. and with 5 °C increase, it will be around 27.5 Mt. (Aggarwal and Swaroopa Rani, 2009). In order to keep pace with the increasing population, India needs to increase its wheat production. In a study done by International Maize and Wheat Improvement Center (CIMMYT) the areas with lower potential productivity will have more negative impact and in future with global warming, their productivity will further decrease under heat stress condition (Joshi et al., 2007). All of these findings indicate that there is a need to understand the effects of rising temperatures on wheat growth and adoption of climate smart practices (Niles et al., 2015). It means that if no mitigation efforts are taken to lessen the emission of greenhouse gases, crop production will be negatively affected to a great extent. Hence, adaptation strategies towards climate change are needed urgently to minimize climate change impacts (Niles et al., 2015). Some of the strategies to reduce impact of terminal heat stress on wheat can be breeding of more tolerant cultivars, conservation agriculture, enhancing water use efficiency and rescheduling the sowing dates. In this study we have taken three agronomic approaches as an adaptation strategy for mitigation of terminal heat stress viz. cultivars mainly cultivated in the area, change of sowing date to matching phenology of wheat crops and alteration in fertilizer rate as well as giving a supplemental irrigation to know how it affects wheat yield under heat stress condition. Crop simulation model- Info Crop is used for studying changes in crop production with respect to change in climatic variables (Aggarwal et al., 2006a). This model can simulate the effect of weather, soil, agronomic management practices, pests, soil carbon, yield, irrigation and greenhouse gas emissions (Hebbar et al., 2008). There are limited studies towards a comprehensive evaluation of the impact of heat stress on growth and yield of wheat crop in the Indo-Gangetic plains of India and its mitigation options. Some studies have been reported on the impact of heat stress on wheat yield however they are mainly based on simulation modelling and lack actual field experiment. Therefore, present study was conducted with both field experiment as well as simulations study with objectives to evaluate the impact of heat stress on growth and yield of wheat crop in the Indo-Gangetic plains of India, to identify different agronomic adaptation options under future climate change scenario for improving heat stress tolerance in wheat crop and to assess economic feasibility as package of practices for adaptation to terminal heat stress in wheat.

2. Materials and methods

2.1. Model description

The impact assessment was done using Info-Crop model. It is a Decision Support System (DSS) written in Fortran Simulation Language (FST) and designed to simulate the effects of weather, soils, agronomic management (including planting, nitrogen, residues and irrigation), and major pests on crop yield and its associated environmental impacts. The input required and output given by model has been given in Table 1. Before simulation study, the model was calibrated for simulating impact of different climate scenario and adaptation strategies and validated by field experiments. Reason for choosing Info-Crop model over other models like APSIM or DSSAT was, it was able to satisfy the growing requirement of assimilated information on climate change, greenhouse gas emission, soil carbon and nitrogen changes. Moreover, the Infocrop model takes into account the loss incurred due to pest and diseases, mostly common in tropical and sub-tropical environment (Aggarwal et al., 2006b). This model has been developed in Indian Agricultural Research Institute (IARI) and the work has been done at IARI research farm which makes use of this model more apt for the given environmental conditions.

2.2. Calibration and validation of model

2.2.1. Calibration of Info-Crop model

Two-year field experiments were conducted during 2011–12 and 2012–13 at IARI research farm, New Delhi with wheat crop. Three different wheat varieties were grown on three different sowing dates under three irrigation schedules (Table 2).The climate of the study area is subtropical, semi-arid with dry hot summer and cold winter. The soil of the experimental site belongs to alluvial group, which is *Typic Ustochrept* with pH 8.6 and sandy loam in texture. From the first year's experimental data T2, T5 and T8 treatments were used for calibration of the model. Observation on phenology, LAI (leaf area index), partitioning of dry matter and yield were taken. Crop coefficients for wheat were calculated by using information from previous published information and the methods followed by Aggarwal and Kalra (1994) and Aggarwal et al., 2006a).Calibration of the varietal coefficients was done from field experiment data. These coefficients were then used in the succeeding validation and application.

2.2.2. Validation of the Info-Crop model

Validation in modelling is done to ensure that calibrated model resembles closely to the actual field condition. It consists of a comparison between model output and actual field data. For the validation

Table 2
Treatment details.

T1	V1 D1 I1	T10	V2 D1 I1	T19	V3 D1 I1
T2	V1 D1 I2	T11	V2 D1 I2	T20	V3 D1 I2
T3	V1 D1 I3	T12	V2 D1 I3	T21	V3 D1 I3
T4	V1 D2 I1	T13	V2 D2 I1	T22	V3 D2 I1
T5	V1 D2 I2	T14	V2 D2 I2	T23	V3 D2 I2
T6	V1 D2 I3	T15	V2 D2 I3	T24	V3 D2 I3
T7	V1 D3 I1	T16	V2 D3 I1	T25	V3 D3 I1
T8	V1 D3 I2	T17	V2 D3 I2	T26	V3 D3 I2
T9	V1 D3 I3	T18	V2 D3 I3	T27	V3 D3 I3

V1: Wheat variety HD 2932; V2: Wheat variety WR 544; V3: Wheat variety HD 2967
D1: Sowing date 30.11.2011 (1st year), 23.11.2012 (2nd year).
D2: Sowing date 9.12.2011 (1st year), 7.12.2012 (2nd year).
D3: Sowing date 19.12.2011 (1st year), 22.12.2012 (2nd year).
I1: Forecast based irrigation (6 cm each); I2: 4 irrigations (6 cm each); I3: 6 irrigations (4 cm each).
1st year: 2011–12; 2nd year: 2012–13.

part, second year's experimental data T2, T5, T8, T11, T14, T17, T20, T23 and T26 treatments were used. Observations on phenology, LAI, partitioning of dry matter and yield were taken. All the simulated growth parameters from phenology to yield were compared with observed experiment data sets.

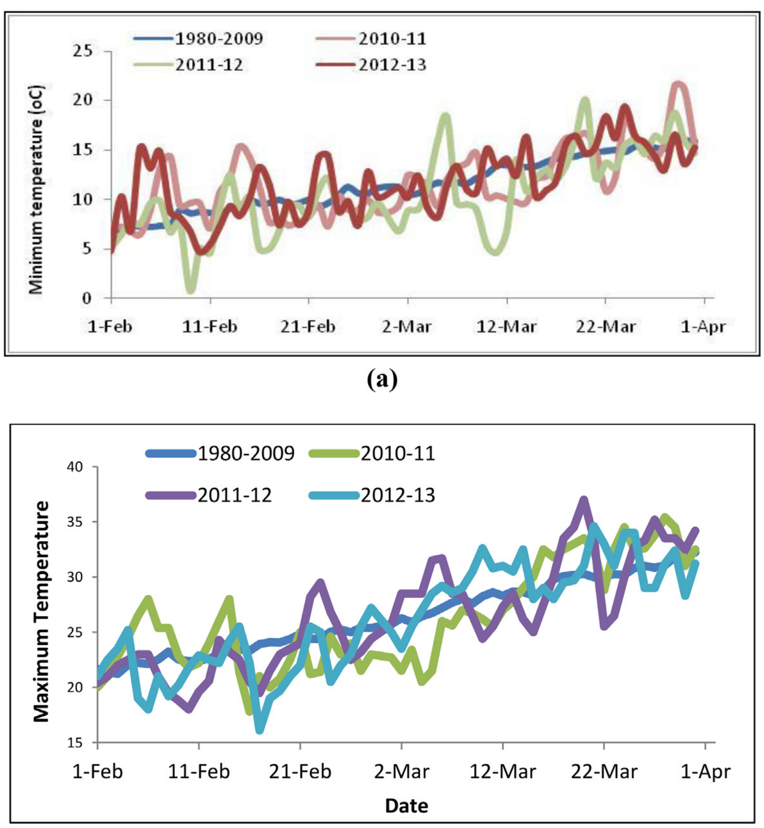
2.3. Simulating the impact of climate change on wheat growth and yield

The climate scenarios developed covered a wide range of predicted changes. Firstly, temperature data for a period of 30 years (1980–2009) were analyzed for Indo-Gangetic plains in relation to terminal heat stress for wheat. Temperature study was done for February to March period in which terminal heat stress period coincides with wheat grain filling duration in India. Fig. 1 (a and b) shows the mean maximum and minimum trend of temperature during last 30 years during February and March. Trend analysis reveals that year 2010–11 was one of the most heat stressed period for wheat crop where more than 14 days were above the optimum temperature (> 31° C) during grain filling stage, while year 2011–12 and 2012–13 did not face heat stress during grain filling stages (Fig. 2 a and b).

Impact of projected climate change scenarios on wheat crop was assessed by running and validating the model for Indo-Gangetic plains for 2020 and 2050 scenarios. The projected weather parameters for terminal heat stress (HS) study in wheat were temperature, carbon dioxide (CO_2) and rainfall, which were identified as per IPCC (2014) scenarios for 2020 and 2050.The projected temperature rise as per IPCC, 2014 for year 2020 and 2050 for Indian sub-continent is +1.4 °C and + 2 °C while rainfall was taken as +5% to +10%, respectively. Atmospheric CO_2 concentration of 415 ppm and 490 ppm was taken for year 2020 and 2050, respectively. Impact of change in CO_2, temperature and rainfall in future scenarios on wheat crop was assessed for both heat stress and non-heat stress (normal year). Year 2010 was taken as the base year for heat stress while year 2011 and 2012 were the base year for non-heat stress year (normal year) simulation.

2.4. Analysis of adaptation strategies

Field experiment with different iterations of agronomic adaptation options were carried out for their impact on alleviation of terminal heat stress impact with respect to yield. Of these options, those which have been reported to showed reduction in impact of terminal heat stress namely date of sowing; fertilizer doze and irrigation amount were chosen. Recommended date of sowing in the study area is 15 November



(b)

Fig. 1. Temperature during the terminal stage of wheat growing period during the experiment compared to the long-term trend (1980–2009) in Delhi. (a) minimum temperature (b) maximum temperature.

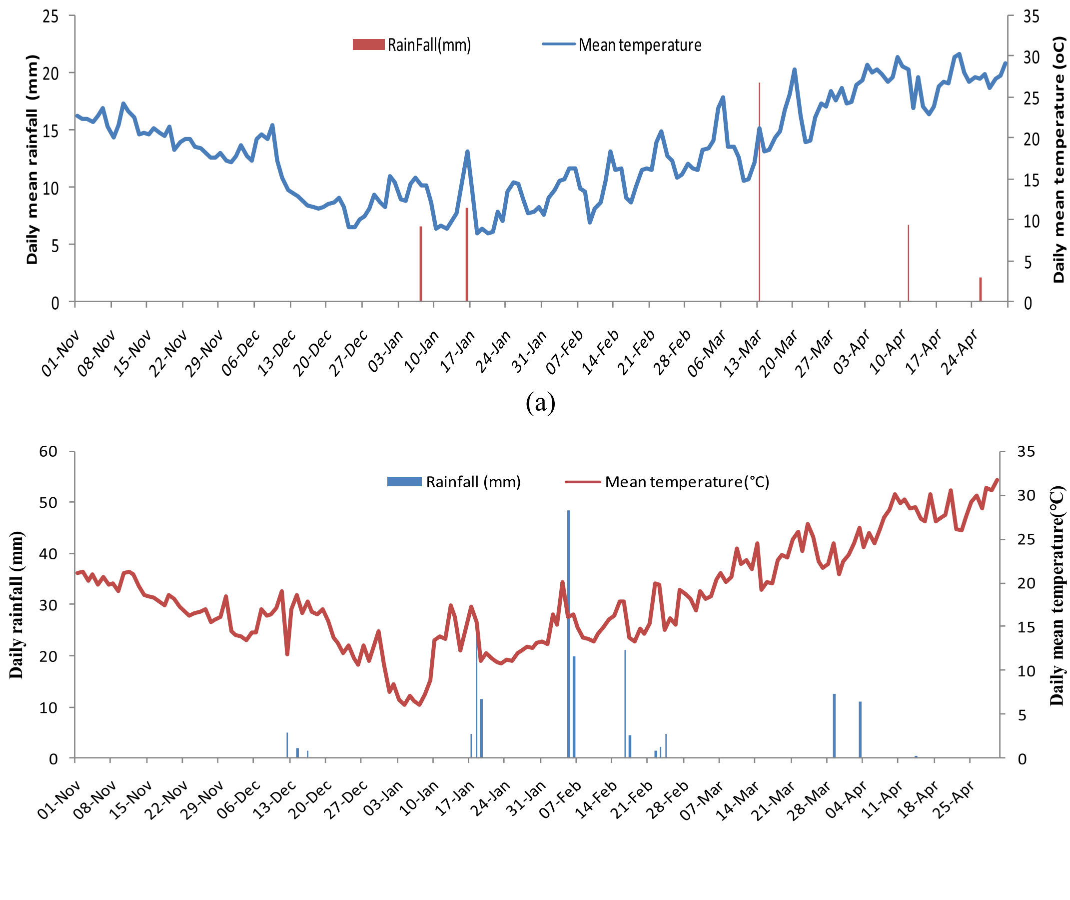


Fig. 2. Temporal variation in mean temperature and rainfall during wheat growing season. (a) 2011–12 (b) 2012–13.

and famer's generally practice to sow the wheat crop near the recommended dates. The recommended dose of fertilizer is 120, 60, 40 Kg NPK and four irrigations given at the sowing, crown root initiation, maximum tillering, and flowering stages. All P and K and a half dose of N are applied at the sowing of wheat. The remaining N is top dressed in two splits at crown root initiation and maximum tillering stages. Prevailing practices at farmers' field are similar to recommended practices (Gupta et al., 2016). Keeping these points in mind following strategies were tested in this study for the adaptation measure.

Advancement in sowing dates by 10 and 15 days compared to normal sowing date (15 November); Application of 30 kg ha^{-1} additional N dose; and supplying a supplemental irrigation at grain filling stage. These adaptation options were tested to analyze the impact of future climate change scenarios on heat stress year. Yield of wheat crop were compared with current management practices and with adaptation measures. Some combinations of these adaptation strategies were also simulated to select the best package of practices to minimize yield loss in wheat.

2.5. Economic feasibility of adaptation options in wheat

Cost of cultivation of wheat under different management practices was calculated by taking into account the cost of field preparation, seeds, fertilizer, manure, irrigation, pesticide and labour charges. Market price of all the inputs as well as hired services on the year of study, were taken into account.

$$\begin{aligned}&\text{Cost of cultivation } (₹\text{ha}^{-1})\\&\quad= \text{Input cost [cost of seed, fertilizer and manures, pesticide}\\&\qquad\text{, diesel and electricity]} + \text{Cost of hiring services}\\&\qquad\text{(labour and machine)]} + \text{other cost (@10\%of total cost)}\end{aligned} \tag{1}$$

Grain yield of wheat was multiplied with the minimum support price of wheat while straw yield with its market price for calculating gross return.

$$\begin{aligned}&\text{Gross return } (₹\text{ha}^{-1})\\&\quad= [\text{Total grain yield (kg ha}^{-1}) \times \text{Minimum support price of grain}\\&\qquad(₹\text{kg}^{-1})] + [\text{Total straw yield (kg ha}^{-1}) \times \text{market price of straw}\\&\qquad(₹\text{ha}^{-1})]\end{aligned} \tag{2}$$

Net income of the farmers was calculated as the difference between gross return and cost of cultivation.

$$\text{Net return } (₹\text{ha}^{-1}) = \text{Gross return } (₹\text{ha}^{-1}) - \text{Cost of cultivation } (₹\text{ha}^{-1}) \tag{3}$$

Benefit cost ratio was calculated by the following formula:

$$\text{Benefit} - \text{Cost Ratio (B: C)} = \text{Gross income } (₹)/\text{Total Cost } (₹) \tag{4}$$

Different adaptation options and their combinations which showed gain in yield under future scenarios were analyzed and compared in terms of their possibility under actual field conditions and economics.

3. Results

3.1. Calibration of model

Info-Crop wheat was calibrated agreeably with precise crop growth parameters. Days to anthesis in observed (74–93) and simulated (76–95) were in agreement with an RMSE of 1.95.Days to physiological maturity in observed ranged from 101 to 148 days and simulated values from 102 to 149 days. For crop under late sown condition the model somewhat predicted towards higher side compared to observed one with RMSE was 1.7. Similarly the simulated LAI at different crop growth stages followed similar pattern as that of measured LAI in all the treatments. Observed and simulated LAI value varied from 2.0–4.2 and 2.2–4.6, respectively with a RMSE of 0.23. Observed and simulated biomass of wheat crop in different treatments ranged from 12.8 to 14.1 t ha^{-1} and 12.5 to 13.9 t ha^{-1} respectively. Predicted biomass by model matched reasonably well with measured observations. RMSE was found to be 0.54. Observed grain yield of HD2932 wheat variety under different sowing dates varied from 2.54 to 5.06 t ha^{-1} while simulated values ranged from 2.57 to 5.09 t ha^{-1} against the RMSE of 0.31.

3.2. Validation of model

The Info-Crop wheat model was able to simulate well the phenology, growth and yield parameter under normal and late sown conditions in different wheat varieties and irrigation schedule. Observed values for days to anthesis (78–97 days) and physiological maturity (108–130 days) matched satisfactorily well with simulated value of days to anthesis (77–95 days) (Fig. 3) and physiological maturity (112–135 days) (Fig. 4). Similarly, observed values of LAI, biomass and grain yield were 1.1–4.6, 10–14.1 t ha^{-1} and 4.2–5.0 t ha^{-1} respectively. While simulated values were 1.0–4.3, 10.1–14.9 t ha^{-1} and 4.1–5.5 t ha^{-1} respectively for LAI (Fig. 5), biomass (Fig. 6) and grain yield (Fig. 7).

To check model performance statistical parameters like Root Mean Square Error (RMSE) (Fox et al., 1999), Coefficient of determination (R^2), D-index (Willmott, 1982), Coefficient of Residual Mass (CRM), and Model efficiency (ME) (Singh et al., 2003) were analyzed using the coefficients developed as given in Table 3. R^2 values between 0.68 and 0.96 depicts that the measured values were in linear agreement with predicted values by model. RMSE ranged from 0.19 to 2.54 in all the cases, which indicate deviation was less in prediction of growth and yield parameters while it showed higher deviation in phenology. D values in the Table 3 suggest that values are closer to 1 which assures that model is simulating well with actual data. CRM values varied from −0.016 to 0.030 showing small error of under and above evaluation. Model efficiency nearer to 1 (0.65 to 0.99) indicating good performance of the model. From the statistical estimates, it was established that the model Info-Crop-wheat can be used efficiently to forecast phenology, growth and yield of wheat crop.

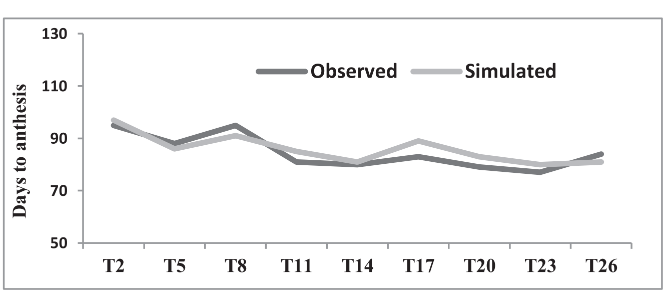


Fig. 3. Effect of sowing dates variety and irrigation levels on observed and simulated values of days to anthesis.

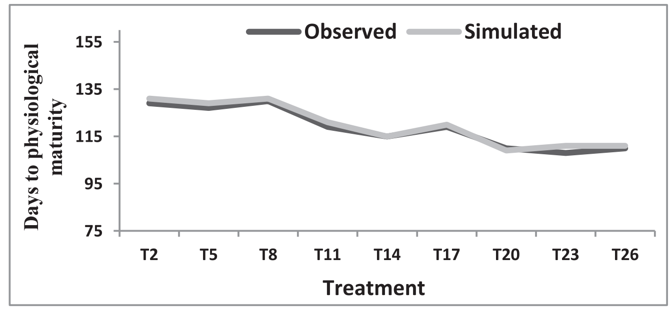


Fig. 4. Effect of sowing dates, variety and irrigation levels on observed and simulated values of days to physiological maturity.

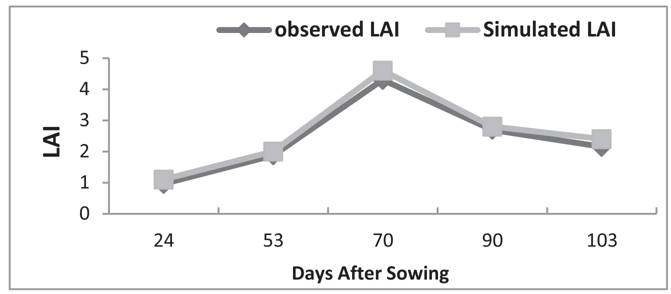


Fig. 5. Effect of sowing dates, variety and irrigation levels on observed and simulated values of leaf area index.

3.3. Impact assessment

Simulated results showed that grain yield of wheat in non-heat stress year was 6.01, 5.92 and 5.85 t ha^{-1} in present, 2020 and 2050 scenarios, respectively. Under heat stress condition yield declined to 4.92, 4.94 and 5.20 t ha^{-1} in present, 2020 and 2050 scenarios respectively (Fig. 8). Percent reduction in grain yield with terminal heat stress condition was 18.1% under present scenario. In scenarios 2020 and 2050 this reduction was 16% and 11.1% respectively (Fig. 8a). Biomass yield followed similar trend to grain yield where reduction was 11.6%, 10.6% and 8.6% in present, 2020 and 2050 scenarios, respectively (Fig. 8b).

3.4. Adaptation options for mitigating terminal heat stress in wheat

3.4.1. Change in sowing date

As a mitigation part re-matching the crop phenology to cropping season can be one such option in which planting dates can be advanced by 10 to 15 days from recommended sowing date. By advancing the planting date from recommended sowing date of 15 November by 10 days to 5 November, yield gain was 0.4%, 3.2% and 3.5% in present, 2020 and 2050 scenarios, respectively (Fig. 9a). While by advancing the sowing dates by 15 days (1st November) yield gain in present, 2020 and 2050 scenarios were 0.8%, 3.4% and 4.4% respectively (Fig. 9b). Thus it was found that under terminal heat stress condition planting wheat crop 10 or 15 days earlier than non-heat stress year date of sowing leads to some yield gain. The gain in yield under present situation was negligible but, in case of future scenarios it had a positive gain of 3.2–3.5% and 3.4–4.4% in 10 and 15 days earlier planting dates in 2020 and 2050, respectively.

3.4.2. Additional dose of nitrogen fertilizer

Under field condition in India in Indo-Gangetic Plains 120 kg of nitrogen (N) is applied in wheat crop during *rabi* season. To combat heat stress, an additional dose of 30 kg N during grain filling stage was tested as an adaptation option. Results revealed that application of an additional N dose of 30 kg ha^{-1} increased grain yield by 3.5%, 2.4%

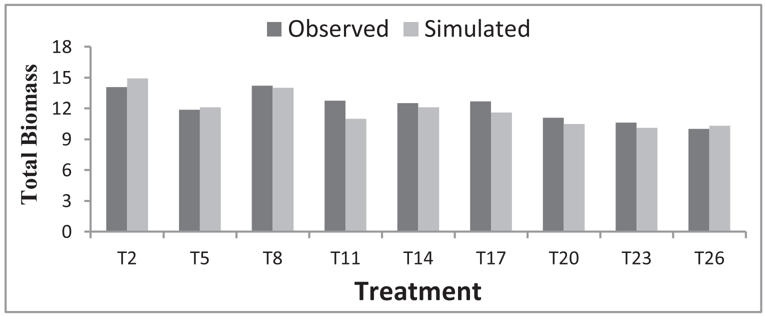


Fig. 6. Effect of sowing dates variety and irrigation levels on observed and simulated values on biomass yield (t ha^{-1}).

and 2.3%, respectively in present, 2020 and 2050 scenarios (Fig. 9c).Yield of wheat crop was found to be same (5.09 t ha^{-1}) with application of 150 kg N (120 + 30 Kg N ha^{-1}) in present and 2020 scenario.

3.4.3. Additional irrigation

An additional irrigation at grain filling stage increased yield by 0.4%,2.2% and 2.9%in present, 2020 and 2050 scenarios respectively (Fig. 9d). In the present scenario grain yield (increased by 0.4% only) did not increased much but under future scenarios of 2020 and 2050% yield increase was found to be higher by 2.2 and 2.9%, respectively.

3.4.4. Combination of adaptation options

Different adaptation options discussed above like date of sowing, N fertilizer application and irrigation were tested in isolation or singly but we tested different combination of adaptation options which may be practical to apply in field and give better yield. After applying different iterations, four best adaptation combinations came out as given below:

a) Advancing sowing date by 10 days (5 November) with 150 kg N
b) Advancing sowing date by 15 days (1 November) with 150 kg N
c) Advancing sowing date by 10 days(5 November) with 150 kg N and 5 irrigations
d) Advancing sowing date by 15 days(1 November) with 150 kg N and 5 irrigations

In condition ‘a’ where the dates of sowing was advanced by 10 days with a combination of an additional 30 kg ha^{-1} N dose with regular irrigation (4 in number) the yield was enhanced by 4.3, 2.8 and 6.5% respectively in present, 2020 and 2050 future climate change scenarios (Fig. 10 a). While the yield enhancement was 5.9, 6.0 and 6.9% in present,2020 and2050 scenarios under ‘b’ where the dates of sowing was advanced by 15 days (Fig. 10b).Under condition ‘c’ when date of sowing was advanced by 10 days with 150 kg N and 5 irrigations the yield enhancement was around 4.5, 2.8 and 6.5%, respectively in present, 2020 and 2050 scenarios while in ‘d’ it was by 7.5, 6.4 and 9%, respectively in present, 2020 and 2050 heat stressed scenarios (Fig. 10c and d).

Table 3
Performance of Info-Crop-wheat model based on different statistical parameters.

Statistical parameters	Days to flowering (days)	Days to maturity (days)	LAI	Biomass (t ha^{-1})	Grain yield (t ha^{-1})
R^2	0.68	0.86	0.96	0.71	0.96
RMSE	2.54	1.67	0.26	0.81	0.19
D index	0.96	0.96	0.98	0.92	0.86
CRM	−0.016	−0.024	−0.038	0.029	0.030
ME	0.84	0.99	0.99	0.65	0.75

3.5. Economic feasibility of different adaptation options

Different adaptation options showed that yield can be increased through one way or other but the best way to know the adoptability of an option at field level is through its economics i.e. it should be economical and easily taken up by the stakeholders. Using this as a base, economic assessment of various adaptation options through Info-crop model was done. For this benefit cost analysis of all best chosen five adaptation options was undertaken (Table 4). Among all, best adaptation option was found to be in 1Nov +150 kg N (advancing planting date by 15 days and additional 30 kg N fertilizer dose). It was well understood as with rise in temperature yield is found to decline so planting the crop early has benefitted wheat crop by completing the grain filling duration before the onset of high temperature: an escaping mechanism; secondly input cost was reduced because one irrigation for soil preparation was not required as plant can use residual soil moisture which helped in reducing input resources and cost. Additionally, providing an extra N dose helped in yield gain since very late top dressing with 30 kg N ha^{-1} after anthesis also helped in enhancing the protein content in grain and hence higher grain yield (Stapper, 2007).

The second best option was 5th Nov + 150 kg N (advancing planting date by 10 days and additional N fertilizer dose). Highest B: C ratio of 2.18, 2.24 and 2.25 was found in present, 2020 and 2050

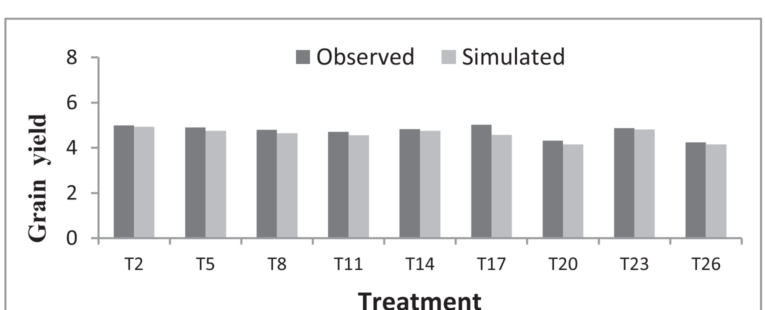


Fig. 7. Effect of sowing dates variety and irrigation levels on observed and simulated values of grain yield (t ha^{-1}).

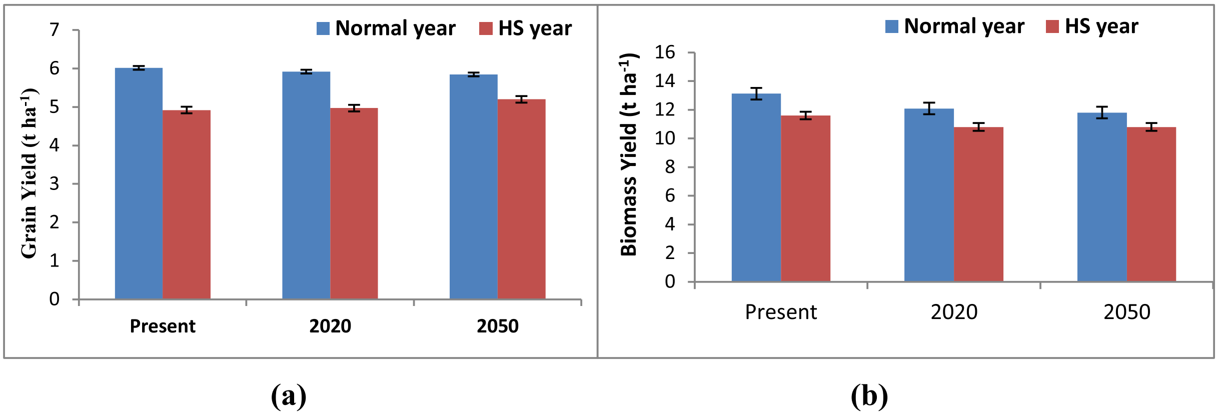


Fig. 8. Simulated percent change in grain and biomass yield in three different scenarios of climate change in wheat.

scenario with adaptation option of '1st November + 150 kg N' as yield was at higher end. Second best adaptation option was '1st November + 150 kg N + 5 irrigations with a B: C ratio of 2.16, 2.07 and 2.22 in present, 2020 and 2050 scenario. Here giving an additional N helped in increased grain weight with irrigation facilities leading to better nitrogen assimilation by plants (Stapper, 2007).

4. Discussion

4.1. Impact assessment

For wheat growth and development optimum temperature lies in between 18° to 24 °C (Ottman et al., 2012). Yield reduced in heat stress period compared to non-heat stress periods and percent yield reduction was less in future scenarios compared to present one. The reason may be as in future climate scenarios increase in CO_2 concentration will increase crop production mainly by stimulating photosynthetic processes and improving water use efficiency leading to increase in the number of productive tillers. But this positive effect of increased CO_2 and yield is nullified by rise in temperature especially in sub-tropical and tropical regions due to increased respiration rates and declined photosynthetic rate resulted in decreased grain filling duration (Pathak et al., 2003). The reduction in photosynthetic rate is concomitant with increase in respiration rate under high temperature condition, which reduces the yield in future climate change scenarios. Similar results were reported by Boomiraj et al. (2010) in mustard where under higher CO_2 concentration yield was enhanced due to changes in physiological processes, but the increase in atmospheric temperature counteracted enhanced dry matter and yield of the crop. A study by Singh et al. (2013) revealed that wheat is one of the thermal sensitive crops.

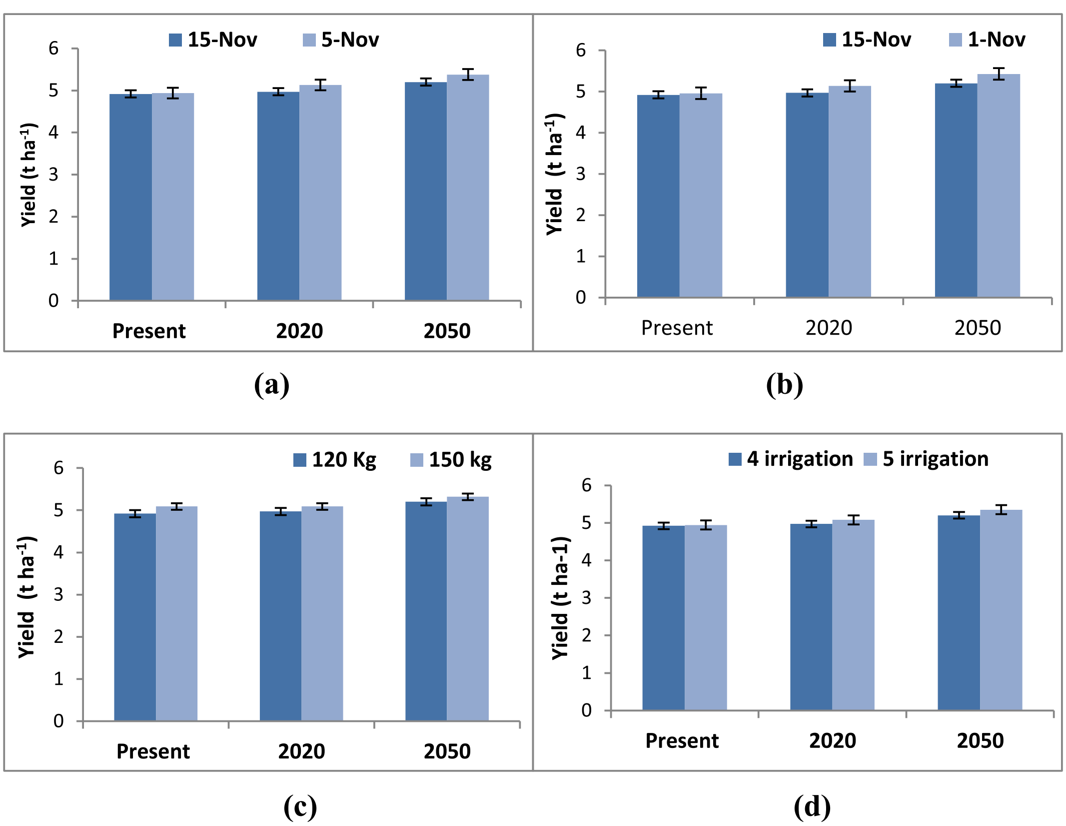


Fig. 9. Impact of various adaptation options for alleviating terminal heat stress on wheat yield; (a) Advancing planting date by 10 days (b) Advancing planting date by 15 days (c) Additional N Fertilizer dose (d) Additional supplemental irrigation.

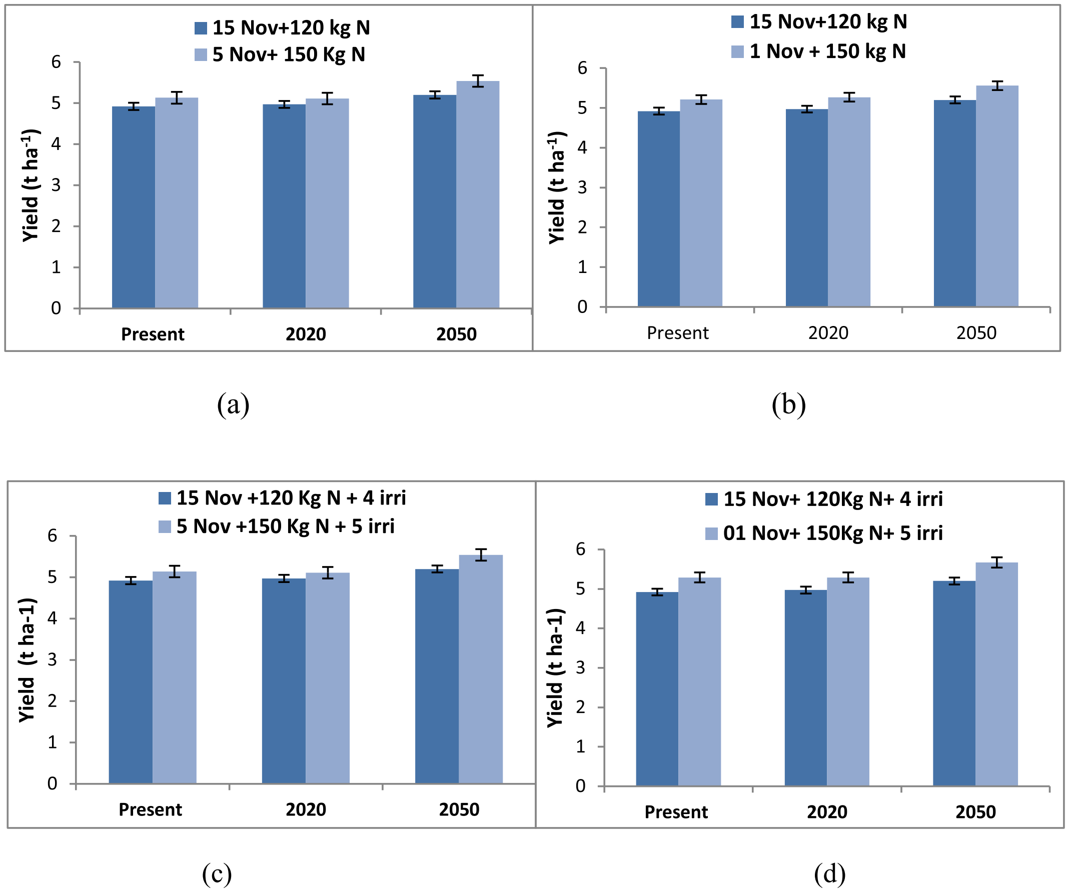


Fig. 10. Impact of various adaptation options combination for alleviating terminal heat stress on wheat yield.

Table 4
Benefit-cost ratio of different adaptation options under terminal heat stress.

Adaptation options	Scenarios	B:C
15 Nov + 120 kg N	Present	1.97
	2020	1.91
	2050	2.00
5 Nov + 150 kg N	Present	2.08
	2020	1.98
	2050	2.16
1 Nov + 150 kg N	Present	2.18
	2020	2.14
	2050	2.25
5 Nov +150 kg N + 5 irrigation	Present	2.02
	2020	1.91
	2050	2.09
01 Nov + 150 kg N+ 5 irrigation	Present	2.16
	2020	2.07
	2050	2.22

Blumenthal et al. (1991) in a long term experiment showed terminal heat stress and grain yield are negatively correlated in wheat crop. Wheat yield decreased with temperature increase above optimum (> 35 °C) affecting leaf chlorophyll, reduced assimilate translocation from source to sink and inhibited starch synthesis in developing grains.

4.2. Mitigation options

To counteract heat stress some options that can be adopted at field level are change in planting date, additional life-saving irrigation and additional N fertilizer dose. When date of sowing was advanced by 10 or 15 days' yield gain was observed as earlier sown crop to some extent able to complete their grain filling duration before the onset of high temperature. This result was in agreement with Mall et al. (2006) who reported that pre-poning sowing dates by 10 days in late and normally sown varieties resulted in higher yield under a modified climate, which was otherwise showing reduced yield. Earlier sown crop escapes heat stress while in late sown condition when temperature increases even for 5–6 days' yield loss is 20% more compared to early sown as observed by Kumar and Rai (2014). Therefore, early planting can be one good option to combat heat stress so that grain filling stage falls under the cooler environment (Farooq et al., 2011). Rice and wheat are the major cereal crops of the Indo-Gangetic plain (IGP). Rice crop is transplanted in the month of July while wheat is sown in mid-November. Early planted wheat in first week of November will be harvested 10 days early in March while succeeding rice crop is planted in July. Hence, early harvesting of wheat may allow growing a third short-duration crop during April and July.

An additional N dose helped in mitigating yield loss due to heat stress condition. Under suboptimal environmental conditions such as high temperature or water stress, it was found that plant performance can be improved by nitrogen fertilizer application (Nunes et al., 1993) This additional N of 30 kg ha^{-1} was applied during grain filling stage in present and future scenarios had positive influence on yield indicated that increased nitrogen supply might have alleviated problems of photo-inhibition (Nunes et al., 1993). Result showed that yield of wheat crop by application of additional dose of N by 30 kg ha^{-1} was same in present and 2020 scenarios. Similar results were found by Dupont et al. (2006) where adding nitrogen fertilizer after anthesis in 37/28 °C diurnal variation did not increased either the rate of protein accumulation or its duration to increase protein level under high temperature regime which was otherwise accumulated under the 24/17 °C regimen hence the leaf senescence earlier in high temperature leading to lesser grain yield. Third option of saving yield loss due to terminal heat stress was application of a life-saving irrigation at grain filling stage which showed yield gain in present as well as future scenarios. Yield gain was negligible in 'present' scenario while future scenarios of 2020 and 2050 had significant gain in yield because in 'present' scenario situation taking CO_2 concentration of 415 ppm, even 1 °C rise in air temperature leads to 37 mm more potential evapo-transpiration and thus fertilization effect of CO_2 is nullified while under future climate change scenarios, CO_2 enrichment with an additional irrigation will increase water use efficiency by less PET resulting from reduced stomatal conductance and increased shading from larger leaves which will ultimately get converted to higher grain yield (Yoshimoto et al., 2005). Similarly, Salvucci and Crafts-Brandner (2004) reported in Pima cotton that those crops having assured irrigation facility could keep their stomata open even at high temperature by means of transpirational cooling. This helps in decreasing the leaf temperature even when temperature difference is more than 10 °C between air and leaf temperature, which results in higher yield by keeping cooler canopies due to increased stomatal control. When date of sowing is advanced and an additional irrigation as well as additional N dose of 30 kg ha^{-1} is provided at grain filling stage, yield enhancement was much higher than by any single adaptation options. The effect can be understood as when crop is early sown they complete their grain filling duration before onset of high temperature thus reducing spikelet sterility, and additionally if we are providing irrigation and N dose at grain filling stage which converts the assimilated starch into filled grains (increasing 1000 grain weight) by reducing canopy temperature depression (less leaf senescence) resulting increased yield of crop (Gupta et al., 2010).

However, early planting of wheat can only become feasible if the rice is harvested early (October end) and land is vacated early. In the Indo-Gangetic plains where rice-wheat cropping system is followed and early planting needs changes in agro-technological management of other crops grown during the remaining part of the year (Wassmann et al., 2009). In case crops cannot be vacated at scheduled recommended time, one can save yield loss by giving an additional N fertilizer and/or providing a supplemental irrigation.

5. Conclusion

Climate change due to elevated CO_2 concentration in the atmosphere would stimulate biomass production in crops particularly C3 crops like wheat. But simultaneously increasing heat stress due to elevated temperature and resulted moisture stress will reduce yield of crops. Sensitivity of wheat towards variations in temperature gets established by Info-Crop wheat model. It confirmed that rise in temperature reduces growth and negatively impacts yield determining parameters in wheat crop. Simulated yield of wheat for terminal heat stress years was found lower compared to non-stress years (normal years). With terminal heat stress years becoming more prevalent than ever, adaptation options for farmers are advancing planting date, selecting more thermo-tolerant cultivars and modifying irrigation and nitrogen-rates. Spikelet sterility (leading to lower yield) caused due to high temperature can be reduced to a certain extent by tuning the dates of sowing so that high temperature stress does not coincide with flowering stage of the crop. Out of various combinations tested for alleviating terminal heat stress in wheat, the best option identified was planting of wheat in first week of November, applying 30 kg additional N ha^{-1} and an additional irrigation at grain filling stage.

(a) Advancing planting date by 10 days +150 kg N-fertilizer
(b) Advancing planting date by 15 days +150 kg N-fertilizer
(c) Advancing planting date by 10 days +5irrigation + 150 kg N-fertilizer
(d) Advancing planting date by 15 days+5 irrigation+150 kg N-fertilizer

Declaration of Competing Interest

None.

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